

Yash Verma

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PROFESSIONAL SUMMARY

- Backend engineer with **4+ years** operating and scaling production infrastructure. At BrowserStack, owned **cloud infrastructure** (multi-region AWS), **CDC pipelines**, and **zero-downtime migrations** on tables with **1B+ rows**, driving **208x** performance gains and hardening production security against critical vulnerabilities. Hands-on with **AWS, Docker, Kubernetes, Kafka, Terraform, Ansible, Prometheus, Grafana, Loki, Linux, Python, Go, and Bash** — comfortable owning systems from design through on-call reliability.

PROFESSIONAL EXPERIENCE

- BrowserStack** Mumbai
Software Engineer - Backend, Platform Team Apr 2025 – July 2026
Infrastructure & Reliability
 - Led the launch of BrowserStack's Australia AWS region (VPC, networking, compute) for GDPR/data-residency, now serving **360K requests/hour** in production.
 - Architected a **CDC pipeline** (35 Debezium connectors) with **Terraform**-provisioned infrastructure, syncing **10TB** MySQL to **ClickHouse** across 4 regions, eliminating manual config drift.
 - Drove **208x query speed up** (9.6s to 46ms) on a 524M-row analytics table via zero-downtime schema migrations across 4 regions.
 - Executed zero-downtime primary-key migrations on 3 production tables of **1B+ rows** (1 TB, 2.9M rows/day) to prevent production outages.
 - Offloaded heavy aggregate queries from MySQL to ClickHouse, cutting database CPU utilization from **96% to 30%** and enabling future capacity growth.*Incident Response & Security Hardening*
 - Resolved **2 critical production vulnerabilities** (RCE and SSRF bypass) end-to-end, from triage and root-cause analysis to fix and rollout.
 - Remediated SQL injection across **200+ APIs** and shipped Static (SAST) CI security scanning covering **98% of production traffic**.
 - Enabled TLS for all database connections and led org-wide **RBAC** rollout (roles, hierarchies, 14 check-points) for **50k+ customers**.
 - Halved per-session auth calls (2 to 1) via a lightweight config service, cutting **102K auth calls/day** and reducing downstream load.
- Zus (formerly Ochain)** Remote
Software Engineer - Backend, Blockchain Storage & Mobile (Contract) Feb 2022 – Mar 2025
 - Automated dynamic Sharder/Miner node management and view-change protocols across a distributed cluster, boosting developer productivity by **90%**.
 - Optimized S3-compatible distributed storage handling **500M+ files** and 40TB+ across **50+ nodes** at sub-100ms latency and **99.9% uptime**.
 - Built a **UAT automation suite** for blockchain storage infrastructure, cutting bugs by **90%** and release cycles from 2 weeks to 3 days.
 - Enhanced multi-wallet concurrency via a unified Go SDK, supporting **10K+ concurrent** transactions and **300%** load-test throughput gains.

PROJECTS

- JOB-BORG**: Multi-user job-search automation platform (**Django, Celery, Redis, PostgreSQL**) with a **Dockerized**, multi-container deployment instrumented end-to-end with **Prometheus, Loki, and Grafana** for metrics, logs, and dashboards.
- Container Runtime Prototype (Docker-like, From Scratch)**: Built an isolated execution environment using Linux **namespaces, cgroups**, layered filesystem concepts, and process sandboxing.
- Load Balancer & TCP Stack (From Scratch)**: Built a TCP/HTTP load balancer (round-robin, least-connections, health checks) plus socket-level **networking** modules with connection lifecycle and flow-control primitives.

OPEN SOURCE CONTRIBUTIONS

- Oppia Foundation (GSoC) [Merged PRs]** Mountain View, CA (Remote)
OSS Contributor & GSoC Jun 2021 – Jan 2022
 - Contributed to **Gradle to Bazel** migration, achieving **10x improvement** in build speeds.
 - Built **CI/CD pipelines** (GSoC) automating build triggers and deployment to **Google Drive, GCP**, and **Play Store** for 4+ tracks.

TECHNICAL SKILLS

Cloud & Infra: AWS, Docker, Kubernetes (K8s), Terraform, Ansible, Helm **Monitoring:** Prometheus, Grafana, Loki

Languages: Go, Python, Java, Bash, SQL, Ruby, Kotlin, C, C++

Databases: MySQL, ClickHouse, PostgreSQL, Redis, MongoDB

DevOps & Tools: Git, CI/CD, Jenkins, GitHub Actions, SAST, CDC

Messaging: Apache Kafka, Event-Driven Architecture

Concepts: Linux Systems Administration, Networking (TCP/IP, Load Balancing), Distributed Systems, Incident Response & Root-Cause Analysis, Zero-Downtime Migrations, Microservices, Query Optimization

ACHIEVEMENTS

- Selected for **Google Summer of Code 2022** (Oppia Foundation).
- Won **top prize** at the **CAMS Hackathon** for developing its Android app.
- Rated **Specialist** on Codeforces (max-rating **1509**) and **5-star** on CodeChef.
- Practised **700+** DSA problems on Leetcode.